

Interview-Style Questions Based on Conditional Statements

1. Check Even or Odd

Question: Determine whether a number is even or odd. **Explanation:** A number is even if it is divisible by 2. Otherwise, it's odd. - **Input:** Number = 6 - **Output:** Even number

- `num= int (input("Enter the No:"))`
 - `if num%2==0:`
 - `print("Even No")`
 - `else:`
 - `print("odd No")`

 - **OUTPUT:**
 - Enter the No: 6
 - Even No
-

2. Divisible by 5 but Not by 10

Question: Check if a number is divisible by 5 but not by 10. **Explanation:** Use modulo (%) to check if the number `% 5 == 0` and number `% 10 != 0`. - **Input:** Number = 25 - **Output:** Satisfy

- `num = int(input("Enter the No:"))`
- `if num%5==0 and num%10!=0:`
- `print("Divisible ")`
- `else:`
- `print("Not Divisible")`

- **OUTPUT:**
- Enter the No: 25

- Divisible
-

3. Biggest Among Two Numbers

Question: Find the biggest number among two. **Explanation:** Use comparison operators (>) to check which number is greater. - **Input:** A = 4, B = 7 - **Output:** Biggest is: 7

- `a= int (input("Enter the No :"))`
 - `b= int (input("Enter the No :"))`
 - `if a>b:`
 - `print("Biggest is :",a)`
 - `else:`
 - `print("Biggest is :", b)`

 - **OUTPUT:**
 - Enter the No : 4
 - Enter the No : 7
 - Biggest is : 7
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4. Smallest Among Two Numbers

Question: Find the smallest number among two. **Explanation:** Use comparison operators (<) to find the smaller value. - **Input:** A = 4, B = 7 - **Output:** Smallest is: 4

- `A = int(input("Enter first number A: "))`
- `B = int(input("Enter second number B: "))`
- `if A < B:`
- `print("Smallest is:", A)`
- `else:`

- `print("Smallest is:", B)`
 - OUTPUT:
 - Enter first number A: 4
 - Enter second number B: 7
 - Smallest is: 4
-

5. Divisible by 2, 3, and 6

Question: Check if a number is divisible by 2, 3, and 6. **Explanation:** If a number is divisible by both 2 and 3, it is also divisible by 6. - **Input:** Number = 18 - **Output:** Satisfy

- `num= int(input("Enter the No:"))`
 - `if num%2==0 and num%3==0 and num%6==0:`
 - `print("Divisible")`
 - `else:`
 - `print("Not Divisible")`
 - OUTPUT:
 - Enter the No: 18
 - Divisible
-

6. Voting Eligibility

Question: Check if a person is eligible to vote ($\text{age} \geq 18$). **Explanation:** A person is eligible to vote if their age is 18 or above. - **Input:** Age = 19 - **Output:** Eligible to vote

- `age = int(input("Enter the Age:"))`
- `if age>=18:`

- `print("Eligible To Vote")`
- `else:`
- `print("Not Eligible To Vote")`

- OUTPUT:
 - Enter the Age:19
 - Eligible To Vote
-

7. Student Pass/Fail Based on All Subjects ≥ 35

Question: Check if a student passed all subjects (maths, physics, chemistry).

Explanation: Student passes only if marks in all subjects are 35 or more. - **Input:** Maths = 40, Physics = 36, Chemistry = 30 - **Output:** Fail

- `maths = int(input("Enter Maths marks: "))`
- `physics = int(input("Enter Physics marks: "))`
- `chemistry = int(input("Enter Chemistry marks: "))`
- `if maths  $\geq$  35 and physics  $\geq$  35 and chemistry  $\geq$  35:`
- `print("Pass")`
- `else:`
- `print("Fail")`

- OUTPUT:
 - Enter Maths marks: 40
 - Enter Physics marks: 36
 - Enter Chemistry marks: 30
 - Fail
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8. Student Pass if Passed Any One Subject (≥ 35)

Question: Check if the student passed at least one subject. **Explanation:** Use logical OR to check if any one subject has marks ≥ 35 . - **Input:** Maths = 20, Physics = 38, Chemistry = 25 - **Output:** Pass

- `maths= int(input("Maths Marks:"))`
 - `physics = int(input("Physics Marks:"))`
 - `chemistry = int(input("Chemistry Marks:"))`
 - `if maths  $\geq$  35 or physics  $\geq$  35 or chemistry $\geq$ 35:`
 - `print("Pass")`
 - `else:`
 - `print("Fail")`

 - **OUTPUT:**
 - Enter Maths marks: 20
 - Enter Physics marks: 38
 - Enter Chemistry marks: 25
 - Pass
-

9. Student Pass if Passed Any Two Subjects

Question: Check if the student passed any two out of three subjects. **Explanation:** Use a counter or logical conditions to verify two subjects ≥ 35 . - **Input:** Maths = 40, Physics = 20, Chemistry = 36 - **Output:** Pas

- `maths = int(input("Maths Marks:"))`
- `physics = int(input("Physics Marks:"))`
- `chemistry = int(input("Chemistry Marks:"))`
- `if (maths  $\geq$  35 and physics  $\geq$  35) or \`

- `(maths >= 35 and chemistry >= 35) or \`
- `(physics >= 35 and chemistry >= 35):`
- `print("Pass")`
- `else:`
- `print("Fail")`
- OUTPUT:
- Maths Marks: 40
- Physics Marks: 20
- Chemistry Marks: 36
- Pass

10. Biggest Among Three Numbers

Question: Find the biggest number among three. **Explanation:** Compare each pair of numbers using if-else conditions. - **Input:** A = 7, B = 4, C = 9 - **Output:** Biggest is: 9

- `a= int (input("A Number :"))`
- `b= int (input("B Number:"))`
- `c= int(input("C Number:"))`
- `if a>b and a>c:`
- `print("Biggest is:", a)`
- `elif b>a and b>c:`
- `print("Biggest is :",b)`
- `else:`
- `print("Biggest is:", c)`
- OUTPUT:

- A Number : 7
- B Number: 4
- C Number: 9
- Biggest is: 9

11. Smallest Among Three Numbers

Question: Find the smallest number among three. **Explanation:** Use comparison logic to determine the minimum value. - **Input:** A = 7, B = 4, C = 9 - **Output:** Smallest is: 4

- `a= int (input("A Number :"))`
- `b= int (input("B Number:"))`
- `c= int(input("C Number:"))`
- `if a<b and a<c:`
- `print("Smallest is:", a)`
- `elif b<a and b<c:`
- `print("Smallest is :",b)`
- `else:`
- `print("Smallest is:", c)`

- **OUTPUT:**
 - A Number : 7
 - B Number: 4
 - C Number: 9
 - Smallest is : 4
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12. Perfect Square or Not

Question: Check if a number is a perfect square. **Explanation:** A number is a perfect square if the square of its square root equals the number. - **Input:** Number = 49 - **Output:** Perfect square

- `num = int (input("Enter the No:"))`
 - `sqrt= int(num**0.5)`
 - `if sqrt *sqrt ==num:`
 - `print(num, "Is Perfect Square")`
 - `else:`
 - `print(num,"Is Not Perfect Square")`

 - **OUTPUT:**
 - Enter the No: 49
 - 49 Is Perfect Square
-

13. Cars Required for Members (Max 5 per car)

Question: Calculate how many cars are needed for a given number of people.

Explanation: Divide total people by 5 and round up using ceiling logic. - **Input:** Members = 17 - **Output:** Cars needed = 4

- `# Cars Required for Members`

- `members = int(input("Enter Total Members: "))`

- `cars = members // 5`

- `# If there are remaining members, one more car is needed`
- `if members % 5 != 0:`

- `cars = cars + 1`
- `print("Cars needed =", cars)`
- OUTPUT 1:
-
- Enter Total Members: 17
- Cars needed = 4
-
- OUTPUT 2:
-
- Enter Total Members: 10
- Cars needed = 2
-
- OUTPUT 3:
-
- Enter Total Members: 23
- Cars needed = 5

14. Second Biggest Among Three Numbers

Question: Find the second largest number among three inputs. **Explanation:** Use sorting or nested conditions to find the second largest value. - **Input:** A = 10, B = 25, C = 18 -

Output: Second biggest: 18

- `a= int (input("A Number :"))`
- `b= int (input("B Number:"))`
- `c= int(input("C Number:"))`
- `list1= [a,b,c]`
- `list1.sort()`
- `print("Second biggest:", list1[1])`

- OUTPUT:
- A Number : 10
- B Number: 25
- C Number: 18
- Second biggest: 18

15. Leap Year or Not

Question: Check if a given year is a leap year. **Explanation:** A year is a leap year if it is divisible by 4, and (not divisible by 100 unless divisible by 400). - **Input:** Year = 2024 -

Output: Leap year

- `# Leap Year or Not`
-
- `year = int(input("Enter the Year: "))`
-
- `if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):`
- `print("Leap Year")`
- `else:`
- `print( "Not a leap year")`
-
- OUTPUT 1
- Enter the Year: 2024

- 2024 Leap Year

- OUTPUT 2

- Enter the Year: 2023
- 2023 Not a leap year

- OUTPUT 3

- Enter the Year: 2000
- 2000 Leap Year

- OUTPUT 4

- Enter the Year: 1900
- 1900 Not a leap year